

CSCI 5930 01

Daniel Shown

2026-08-19

Syllabus

Course

- **Course Number:**
CSCI 5930 01
- **Course Name:**
Special Topics: Software Product Engineering Leadership
- **Schedule:** Mondays 4:30-5:50 P.M. (joint with Capstone) Wednesdays 4:30-5:50 P.M.
- **Location**
 - **Collocated on Mondays:** BSC 253
 - **Collocated on Wednesdays:** Ritter Hall 115
 - **Virtual:** <https://canvas.slu.edu/courses/97306>
 - **Remote:** Consult if remote options are possible; Additional requirements must be met when working remotely.
- **Description:** A two-semester sequence immerses students in the practice of technical leadership within open source software product development. Each student will serve as a technical lead, guiding an undergraduate capstone team through the creation of a software product tailored to address research or community requirements. This course focuses on technical leadership, team dynamics, product management, agile methods, strategic considerations, and adherence to open source software standards.

Instructor

- **Name:** Daniel Shown
- **Pronouns:** He/him/his
- **Title:** You do not need to use a title, and I prefer simply “Daniel”. If you are feeling especially formal, then you may also refer to me as “Daniel Shown”. I am looking to develop colleagues, and I want you to feel comfortable speaking to me as you would any other professional colleague.
- **Office hours:** By appointment in-person or Zoom
- **Co-working Space:** SSE 210
- **Office location:** MDH 2012
- **Schedule an appointment:** <https://calendly.com/daniel-shown/30min>
- **Instructor communication:** daniel.shown@slu.edu
- **GitHub User:** <https://github.com/kungfuchicken>

This syllabus was developed with assistance from Claude AI (Anthropic, 2025) for content organization, pedagogical feedback, and clarity refinement.

Course Objective

Deliver valued software for a client by leading a software development team, serving as the primary maintainer of a long-term open source codebase, and building an open source community around the solution.

Learning Outcomes

- Developing
 - Technical leadership principles and practices.
 - Manage core developer team collaborative dynamics.
 - Research, design, implement, and evolve effective solutions.
- Open Source
 - Articulate the value of open source principles.
 - Implement open source required and good practices.
 - Build open source communities and governance.
- Software
 - Apply software engineering principles.
 - Design sustainable software architectures.
 - Build and enhance solutions ready for production use.
- Products
 - Develop and articulate a compelling product vision.
 - Adapt a product vision into achievable project scopes.
 - Gather and respond to project health and changing requirements.

Attendance

I expect you to attend class regularly, on time, and to participate in the class activities. We will have several activities during class time: discussion on course relevant topics, reviewing each other's work, participating in team retrospectives, and practicing for end of semester presentations.

Missing class is highly discouraged. If you miss class, you will be unable to participate in the team activities, thus you will get a reduced grade in that category. If you need to miss class due to an emergency, it is your responsibility to follow-up with your team about the discussion you missed in class and any information the instructor provided.

I recognize that students have diverse needs and circumstances that may affect attendance. If you anticipate attendance challenges due to health conditions, family responsibilities, work obligations, or other circumstances, please reach out early in the semester so we can discuss accommodations. Students with documented accommodations should connect with me to ensure their needs are met. The key to success is proactive communication and collaborative problem-solving.

Materials, and/or Equipment

Core Texts (Choose Your Path):

- **Team Leadership:**
 - **General team dynamics:** *The Ideal Team Player: How to Recognize and Cultivate The Three Essential Virtues*, Patrick M. Lencioni. John Wiley & Sons, 2016. <https://www.tablegroup.com/product/ideal-team-player/>
 - **OR**
 - **Technical team dynamics:** *Debugging Teams*, Ben Collins-Sussman, Brian Fitzpatrick. O'Reilly Media, Inc. 2015 <https://www.debuggingteams.com/>.
- **Open Source:**
 - **Community Building:** *The Art of Community: Building the New Age of Participation (2nd. ed.)*. Jono Bacon. O'Reilly Media, Inc., 2012. <https://www.jonobacon.com/books/artofcommunity/>
 - **OR**
 - **Community Governance:** *The Open Source Way 2.0*. Karsten Wade, et. al. <https://www.theopensourceway.org/>

- **Engineering Wisdom:** *The Art of Doing Science and Engineering*. Richard W. Hamming. Stripe Press, 2020. <https://press.stripe.com/the-art-of-doing-science-and-engineering>
- **Product Development:** *Continuous Discovery Habits: Discover Products That Create Customer Value and Business Value*. Teresa Torres. Product Talk, LLC, 2021. <https://www.producttalk.org/2021/05/continuous-discovery-habits/>

- Additional readings and a schedule for discussions will be posted in Canvas.

Recommended: Choose one book from each category. Students should select readings based on their experience level and project needs. Discussion topics will accommodate multiple perspectives and sources.

Note: You can find the books as hard copy for about \$60, or there are free or electronic versions for significantly less.

The following resources will be used throughout the semester:

- Canvas LMS
- GitHub
- Slack
- Laptop

Optional Resources

There are multiple options for tools addressing these functional needs. Please find a solution that works for you and your teams(s).

- Conceptual Diagrams
- Technical Diagrams
- Integrated Development Environment

Course Structure

This course has two integrated components. The **primary component** involves leading a software development team to deliver value with open source software for a real client. The **secondary component** develops leadership capacities essential for technical leadership in both client-driven and open source contexts.

Students serve as **Tech Leads** for cross-functional teams including a client, undergraduate developers, and potentially industry fellows and external contributors. Success requires applying leadership concepts to guide team dynamics, technical decisions, and product evolution.

Development occurs over **seven two-week sprints**. Each sprint:

- **All team members** produce demo videos of their individual contributions
- **All team members** collaborate on creating new issues for the next sprint
- **Tech Leads** review and merge team pull requests
- **Tech Leads** engage with team demo videos (watch, provide feedback)
- **Tech Leads** ensure sprint planning is completed and take ultimate responsibility for deliverables
- **Tech Leads** develop focused planning documentation

The semester culminates in **product and community strategy documents** that capture the evolving vision and enable effective handoff to future Tech Leads.

Students will engage with concepts, tools, and techniques related to team dynamics, software engineering, product management, and open source community building throughout the semester.

Planned Schedule

Week	Key Class Date	Sprint	Focus Topic	Checkpoint Artifact
0	August 19	0	Pre-planning	
1	August 24	0	Project Pitches	
1	August 26	0	Virtual Onboarding	
2	August 31	0	Issue Readiness	
2	September 2	0	Issue Readiness	
3	September 7	1	Labor Day (No Class)	
3	September 9	1	Teams	Working Agreements
4	September 14	1	Retrospectives	
4	September 16	1	Engineering	Code Review
5	September 21	2	Product	
5	September 23	2	Wellness Day (No Class)	Team Outing
6	September 28	2	Open	
6	September 30	2	Open	Community Building
7	October 5	3	Teams	Retrospectives
7	October 7	3	Teams	Team Metrics
8	October 12	3	Engineering	Milestone 1
8	October 14	3	Engineering	Usability Research
9	October 19	4	Product	
9	October 21	4	Product	Market Evaluation
10	October 26	4	Open	
10	October 28	4	Open	Team Metrics
11	November 2	5	Teams	
11	November 4	5	Teams	Strategy Draft
12	November 9	5	Engineering	
12	November 11	5	Engineering	Threat Model
13	November 16	6	Product	
13	November 18	6	Product	Backlog Audit
14	November 23	6	Open	Community Strategy
14	November 25	6	Open	Thanksgiving Break (No Class)
15	November 30	Launch	Launch Preparation	Milestone 2
15	December 2	Launch	Launch Preparation	
16	December 7	Launch	Launch Day, 4:00-5:50 P.M.	

Note: Subject to change depending on needs and circumstances that emerge.

Evaluation and Grading

Grade Components

Student performance will be evaluated across five key areas that reflect the multifaceted nature of technical leadership:

- **30% Managing Team Dynamics** - Leading collaborative development processes, facilitating client interactions, conducting effective retrospectives, and fostering productive team relationships
- **30% Software Engineering Quality** - Maintaining high code standards, implementing quality engineering practices, conducting thorough code reviews, and establishing sustainable development processes
- **15% Product and Project Management** - Delivering compelling product pitches, executing effective sprint planning, and developing comprehensive product strategy documentation
- **15% Implementing Open Source Practices** - Ensuring OSI compliance, building community engagement, establishing governance structures, and fostering external contributions
- **10% Leadership Capacity Building** - Demonstrating professional leadership through publicly visible activities that contribute to the broader software development community

Note: Weekly PR reviews, planning sessions, and sprint retrospectives contribute to both Managing Team Dynamics and Software Engineering Quality. Checkpoint documents span multiple evaluation areas.

Leadership Capacity Building Requirement

Leadership Capacity Building activities are required for A-level performance and must demonstrate publicly visible professional technical leadership. Students may pursue external contributions (conferences, mentoring, industry blog posts) or contribute to the Open Source with SLU community through collaborative documentation, shared resources, and cross-team knowledge sharing. Examples include:

- **Team Dynamics:** Contributing to project blog posts about team processes, speaking about collaborative development practices, or mentoring other technical teams
- **Software Engineering:** Making meaningful contributions to external open source projects, writing technical blog posts, or presenting at conferences about engineering practices
- **Product Management:** Creating and sharing strategic planning frameworks, writing about product development processes, or contributing to product management communities
- **Open Source:** Building community engagement initiatives, contributing to OSS governance discussions, or creating resources for new contributors, Contributing to OSS blog posts about significant project accomplishments

Detailed requirements and additional examples available in Canvas.

Additionally, students should participate as developers in a short 2 sprint instructor-led collaborative project near the start of the semester to build empathy and understanding of team member perspectives.

Assessment Approach

Evaluation will be based on **professional judgment** to model real-world development experiences and allow flexibility across diverse project types. Assessment will incorporate:

- Instructor observation and review of work products
- Peer feedback from team members and other Tech Leads
- Client input on collaboration and delivery effectiveness
- Evidence of community engagement and professional growth

Students are expected to document their leadership activities through posts on personal websites, professional networking platforms (LinkedIn, etc.), or contributions to project blogs and repositories.

Grading Scale

- **93% and higher:** A
- **90% and higher:** A- or better
- **87% and higher:** B+ or better
- **80% and higher:** B or better
- **77% and higher:** C+ or better
- **70% and higher:** C or better
- **60% and higher:** D or better
- **Below 60%:** F

Norms and Policies

Course Norms

Norm	Do	Avoid
Be Scholarly	Use proper language, grammar and spelling. Be explanatory and justify your opinions. Credit the ideas of others through citing and linking to scholarly resources.	Misinforming others when you may not know the answer. If you are guessing about something, clearly state that you do not know the answer.
Be Respectful	Respect privacy, diversity and opinions of others. Communicate tactfully and base disagreements on scholarly ideas or research evidence.	Personal attacks, dismissive language, or disregarding different perspectives and backgrounds.
Be Professional	Always represent yourself well. Be truthful, accurate and run a final spell check. Nurture a professional culture.	Using profanity or participating in hostile interactions. Using slang and emoticons in formal communications.
Be Polite	Address others by name or appropriate title and be mindful of your tone. Treat people as if you were in a face-to-face situation.	Using sarcasm, being rude or writing in all capital letters. Written words can be easily misinterpreted as they lack nonverbals.

Course Policies

Course Changes Students can request course modifications through discussion with the instructor. Changes affecting the entire class require consensus among all students.

Late or Missing Work This course requires tight coordination of schedules and timely completion of all work. No late work will be accepted for credit on any assignments. Completing missing work may be required to complete subsequent work. Extensions may be granted for documented emergencies or health issues. Communicate with the key stakeholders, including the instructor, client, and development team, as early as possible when circumstances arise.

Team Conflicts Interpersonal issues should first be addressed within the team using professional conflict resolution practices. If resolution cannot be achieved, escalate to the instructor for mediation. Document attempts at resolution before escalation.

Client Communication All client interactions must maintain professional standards. Students represent both themselves and the university. Major client communications should be coordinated with the instructor when appropriate.

Open Source Contributions When contributing to external open source projects, follow established community guidelines and standards. Represent the course and university professionally in all public-facing contributions and discussions.

Generative AI Usage Responsible and effective use of Generative AI is encouraged in this class. You are invited to use AI coding assistants (GitHub Copilot, Claude Code, etc.) to accelerate development and AI tools to improve presentations and documentation. When using Generative AI, you are responsible for thoroughly reviewing generated content, ensuring accuracy, and adapting it appropriately. **Disclose AI assistance in significant work products.**

Appendix: University Resources and Policies

Student Success Center

The Student Success Center (SSC) supports students in reaching their goals in and out of the classroom. Providing a variety of resources, the Student Success Center houses both the Center for Accessibility and Disability Resources (CADR) and Academic Support, which includes Tutoring, Supplemental Instruction, University Writing Services, and Student Success Coaching. The Student Success Center is located in the Busch Student Center, Suite 331, and students can make an appointment with any SSC resource via EAB Navigate.

Learn more: <https://www.slu.edu/life-at-slu/student-success-center/index.php>

University Writing Services

University Writing Services offers one-on-one consultations with trained writing consultants who help with everything from brainstorming, outlining, and proposing research questions to documenting sources, revising, and implementing feedback. These consultations are available to both graduate and undergraduate students and can take place in-person, asynchronously, or via Zoom. All appointments can be scheduled through EAB Navigate – Student.

Getting feedback benefits writers at all skill levels on different writing projects including class assignments, conference papers, cover letters, dissertations, group projects, multimedia assignments, personal statements, senior capstone projects, short answer questions on applications, speeches, and theses.

Contact: writing@slu.edu

Learn more: <https://www.slu.edu/life-at-slu/student-success-center/academic-support/university-writing-services/index.php>

University Counseling Center

The University Counseling Center (UCC) offers free, short-term, solution-focused counseling to Saint Louis University undergraduate and graduate students. UCC counselors are highly trained clinicians who can assist with a variety of issues, such as adjustment to college life, troubling changes in mood, and chronic psychological conditions.

To make an appointment: Call 314-977-8255 (TALK), visit the UCC website, or walk into the clinic on the second floor of Wuller Hall.

After-hours support: Press #9 after dialing the clinic number and ask for a mental health specialist.

Wellness and Support

With our Jesuit commitment to *cura personalis*, the University sees your academic success as connected to your health and well-being and provides resources to support your holistic wellness.

All students experience stressors and challenges at some point, and seeking support is both normal and beneficial. Such challenges may be the result of academic concerns (such as those related to particular assignments or content in a course), or they may be more personal in nature (such as concerns related to relationships, mental health, medical issues, loss, identities, alcohol or drugs, housing or food security, finances, or local/world events, among other things).

If you experience these or other difficulties that are impacting your well-being and/or academic work, please consider seeking support from the resources available to you:

- **For questions or concerns related to this course**, please contact me. I am invested in your success and will support your success in the ways I can.
- **For personal wellness resources**, visit the Well-being page of the SLU website for a comprehensive list of available support services.

Crisis Support If you or someone you know is experiencing a crisis: please consult Crisis Support and Warning Signs on the University Counseling Center website or call the University Counseling Center at 314-977-TALK (8255) and press #9 to be connected to a behavioral health nurse 24/7.

Basic Needs Security

Students experiencing food insecurity, housing insecurity, and any other challenges that are impacting their personal and/or academic wellbeing are encouraged to contact the Dean of Students Office for support.

Contact options: - Submit an intake form online - Email: deanofstudents@slu.edu
- Call: 314-977-9378

Students may also communicate directly with their instructors about any challenges they are experiencing to receive support and resource referrals.

Academic Integrity

Academic integrity is the commitment to and demonstration of honest and moral behavior in an academic setting. Since the mission of the University is “the pursuit of truth for the greater glory of God and for the service of humanity,” acts of integrity are essential to its very reason for existence. Thus, the University regards academic integrity as a matter of serious importance.

Academic integrity is the foundation of the academic assessment process, which in turn sustains the ability of the University to certify to the outside world the skills and attainments of its graduates. Adhering to the standards of academic integrity allows all members of the University to contribute to a just and equitable learning environment that cultivates moral character and self-respect.

Full policy: <https://www.slu.edu/provost/policies/academic-and-course/academic-integrity-policy.pdf>

Disability Accommodations

Students with a documented disability who wish to request academic accommodations must formally register their disability with the University. Once successfully registered, students also must notify their course instructor that they wish to use their approved accommodations in the course.

Please contact the Center for Accessibility and Disability Resources (CADR) to schedule an appointment to discuss accommodation requests and eligibility requirements. Most students on the St. Louis campus will contact CADR, located in the Student Success Center.

Contact CADR: - Email: accessibility_disability@slu.edu - Phone: 314.977.3484

Once approved, information about a student’s eligibility for academic accommodations will be shared with course instructors by email from CADR and within Banner. Students who do not have a documented disability but who think they may have one also are encouraged to contact CADR. Confidentiality will be observed in all inquiries.

Title IX

Saint Louis University and its faculty are committed to supporting our students and seeking an environment that is free of bias, discrimination, and harassment. If you have encountered any form of discrimination on the basis of sex, including sexual harassment, sexual assault, stalking, domestic or dating violence, we encourage you to report this to the University.

Important: If you speak with a faculty member about an incident that involves a Title IX matter, **that faculty member must notify SLU’s Title IX Coordinator that you shared an experience relating to Title IX.** This is true even if you ask the faculty member not to disclose the incident. The Title IX Coordinator will then be available to assist you in understanding all of your options and in connecting you with all possible resources on and off campus.

If you are pregnant or experiencing a pregnancy related condition, the Title IX Coordinator can assist you in understanding your rights and options as well as provide supportive measures.

Title IX Coordinator: Anna Kratky (DuBourg Hall, room 36; anna.kratky@slu.edu; 314-977-3886)

Confidential Resources: - University Counseling Center: 314-977-TALK - Anonymous reporting: SLU Integrity Hotline at 1-877-525-5669 or online at [SLU.EDU/INTEGRITYHOTLINE](https://www.slu.edu/about/safety/sexual-assault-resources/index.php)

Resources and policies: <https://www.slu.edu/about/safety/sexual-assault-resources/index.php>